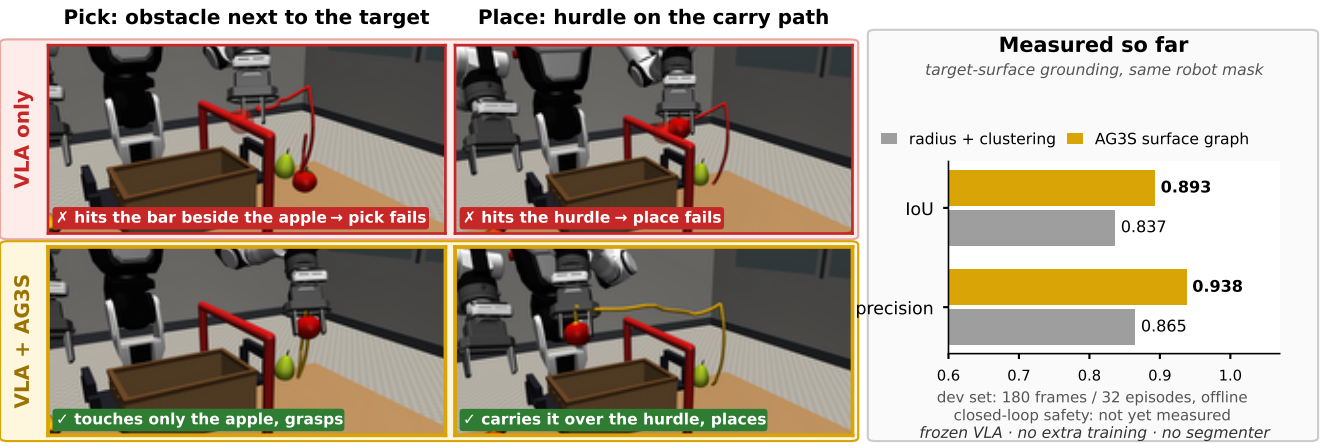

Pick: obstacle next to the target
Place: hurdle on the carry path
VLA only
✗ hits the bar beside the apple → pick fails
✗ hits the hurdle → place fails
VLA + AG3S
✓ touches only the apple, grasps
✓ carries it over the hurdle, places
Measured so far
target-surface grounding, same robot mask
radius + clustering
AG3S surface graph
IoU
0.893
0.837
precision
0.938
0.865
0.6
0.7
0.8
0.9
1.0
dev set: 180 frames / 32 episodes, offline
closed-loop safety: not yet measured
frozen VLA · no extra training · no segmenter
1 Noun attention → seeds
2 Depth → target surface
3 Features → same object?
target 0.61
other fruit 0.11
4 Contact links only